

LECTURE ACTIVITY 10: SLEEPLESS NIGHTS, PART II – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Calculate the standardized statistic under a theory-based approach
- 2) Use a theory-based analysis to conduct a single test involving the mean of a single quantitative variable.
- 3) Identify whether the validity conditions for a one-sample t-test are met.
- 4) Evaluate the strength of evidence against the null hypothesis based on the p-value and/or standardized statistic (t).
- 5) Recognize that even with appropriately gathered and analyzed data, there is a chance of arriving at an incorrect decision (Type I and Type II errors).

Key Terms:

- Theory-Based Inference
- t –distribution
- Central Limit Theorem

Intro / Recap

In the last activity we investigated the following research question: Is the average amount of sleep (in hours) that STAT 1110 students has last night less than the recommended eight hours? The sample we had was composed of 70 STAT 1110 students:

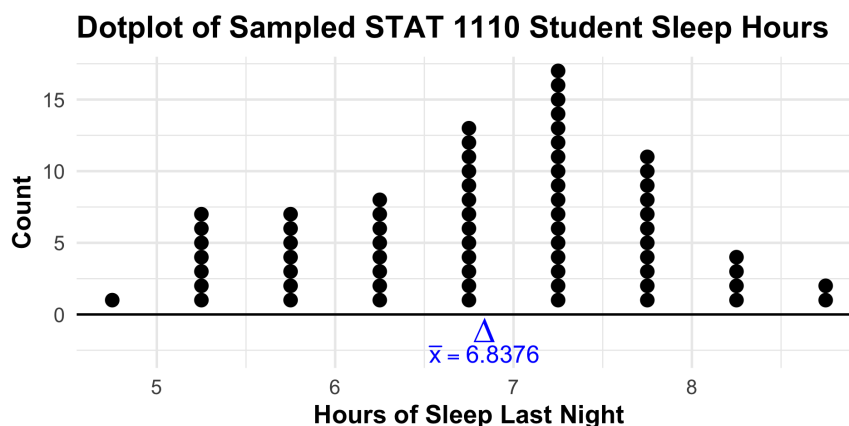


Figure 1: Results from a (fictitious) survey sent to 70 STAT 1110 students.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 6.84$	median = 7	$s = 0.93$	$n = 70$

Table 1: Summary statistics of the (fictitious) data.

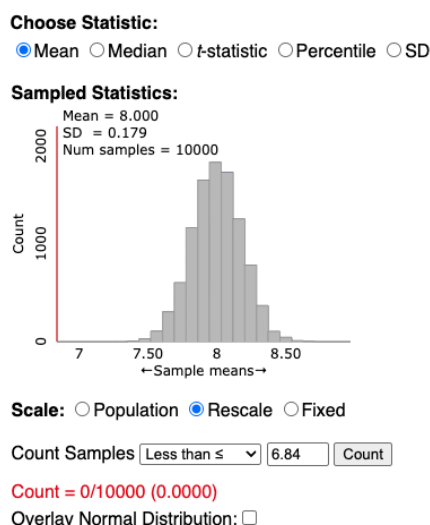


Figure 2: Screenshot from the One Mean Applet

Letting μ denote the true average amount of sleep (in hours) slept by all STAT 1110 students during the night in question, our null and alternative hypotheses are

$$\begin{cases} H_0 : \mu = 8 \\ H_A : \mu < 8 \end{cases}$$

When we conducted a simulation-based test (using the **One Mean Applet**), we obtained an approximate p -value of 0.000, as indicated in Figure (2) to the left.

Another measure of strength of evidence against the null hypothesis that we used was the standardized statistic:

$$\text{Standardized Statistic} = \frac{6.84 - 8}{0.179} \approx -6.48$$

Today, we'll conduct the same analysis using a **theory-based** approach.

Theory-Based Approach (One-Sample T-Test)

In the previous several lecture activities, we saw that we can predict the shape of the null distribution using theory instead of using simulations, as long as certain validity conditions are met. We will now explore a theory-based approach for testing a single population mean.

In Chapter 1 of the textbook, we used a theoretical probability distribution called a **normal distribution** to predict the distribution of sample proportions. In fact, a normal distribution is often a good approximation for the distribution of sample *means* as well.

Concept Overview - Normality

When taking random samples from a large population (more than 20 times the size of the sample), the distribution of sample means will be approximately normal if EITHER of the following hold:

- The population distribution is normal
- The sample size is large (in which case the **Central Limit Theorem** tells us that the distribution of sample means will be approximately normal) **and the sample distribution is not too heavily skewed.**

Pro Tip

Note: if the population is normally-distributed, then we don't need to use the Central Limit Theorem - the distribution of sample mean values will be approximately normal regardless of the sample size.

Additionally, a theory-based approach leverages the following facts:

Concept Overview

When taking random samples from a large population (more than 20 times the size of the sample), the distribution of sample means will:

- Be approximately normal (by the previous Concept Overview)
- Center around the population mean (μ)
- Have a standard deviation smaller than the original population's standard deviation. In particular,

$$SD_{\text{NULL}} = \frac{\sigma}{\sqrt{n}}$$

where σ denotes the population standard deviation and n denotes the sample size.

- Since the value of σ is often unknown in practice, we typically use s as a proxy for it: that is, we use

$$SD_{\text{null}} = \frac{s}{\sqrt{n}}$$

This enables us to construct a general formula for a standardized statistic for a test on a population mean. In particular, a theory-based standardized statistic under the null, here called a **t-statistic**, for a test on a population mean can be found using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where μ_0 denotes the null value, s denotes the sample standard deviation, and n denotes the sample size.



Question 1

Calculate the value of the t-statistic for the class data depicted above. Show your work.

Solution:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{6.84 - 8}{0.93/\sqrt{70}} \approx \mathbf{-10.436}$$

You probably noticed that our reference distribution for this standardized statistic is no longer the normal distribution, but rather the **t-distribution** (and the standardized statistic is often called a **t-statistic**). The shape of a t -distribution is still a mound-shape curve, and it is centered at zero. It looks a lot like a normal distribution, but it is a bit more spread out than a normal distribution (more observations in the 'tails,' less in the middle). The good news is that t -statistics can be interpreted very similarly to z -statistics. Namely, values roughly greater than 2 (or less than -2) can be considered strong evidence against the null hypothesis.

 Question 2

Based on the t -statistic, how much evidence do the sample data provide against the null hypothesis? Choose one.

- ☐ Not much evidence against the null hypothesis
- ☐ Moderate evidence against the null hypothesis
- ☐ Strong evidence against the null hypothesis
- ✓ **Very strong evidence against the null hypothesis**

We can use the **One Mean Applet** to conduct a theory-based test of our hypotheses.

 Question 3

Navigate to the One Mean applet, and:

- 1) Check the “Show Sampling Options” box
- 2) Set the “Number of Samples” to be 10,000
- 3) Set the “Sample Size” to 70
- 4) Click “Draw Samples”
- 5) Above the null distribution change the choice of statistic from “mean” to “ t -statistic”
- 6) Check the box to “Overlay t Distribution”

Don’t close out of the applet yet! Does this theoretical model (the t -distribution) do a reasonable job of predicting the behavior of the simulated distribution of t -statistics? Explain, and include a screenshot below.

Here’s my answer:

Solution: Yes. The t -distribution provides a reasonable model for the simulated distribution of t -statistics because both distributions are bell-shaped, centered around 0, and have similar spreads.

Here’s my screenshot:

Solution:

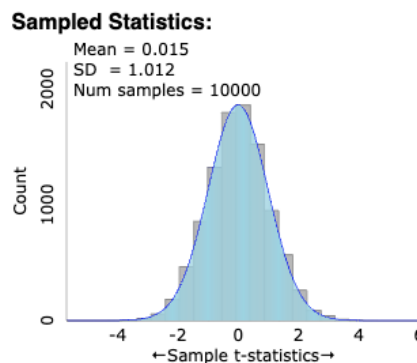


Figure 3: Screenshot from the One Mean Applet.

Question 4

Use the observed t -statistic you found in Question (1) above to find and report the **theory-based p -value** (thereby conducting a **one-sample t -test**). To show your work, **include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) p -value.**

- My theory-based p -value is 0.000

Here's my screenshot:

Solution:

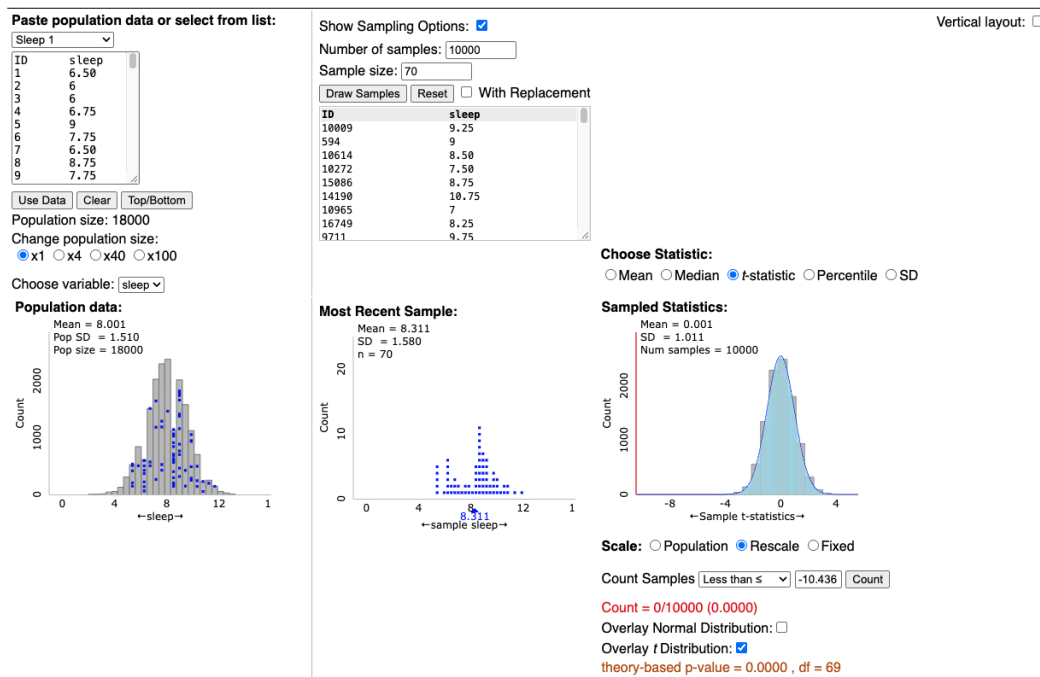


Figure 4: Screenshot from the One Mean Applet.

You should notice that the theory-based p -value is similar to the simulation-based p -value we obtained in the previous lecture activity, using a simulation-based approach. This happened because the **validity conditions** were met.

Concept Overview - Validity Conditions for a Theory-Based Approach

As we discussed in Chapter 1, theory-based tests are valid when certain conditions are met. In the case of a test on a single mean, also called a one-sample t -test, the validity conditions are a little more complicated than they were for a single proportion: you should have **at least 20 observations AND the sample distribution should not be strongly skewed**.

It's important to keep in mind that these conditions are rough guidelines and **not** a guarantee. All theory-based methods are approximations. They will work best when the distributions are symmetric, when the sample sizes are large and there are no large outliers. *When in doubt, use a simulation-based method as a*

cross-check. If the two methods give very different results you should probably consult a statistician.

Question 5

Explain why the validity conditions were met in this study. You may want to reference various pieces of information from earlier in the lecture activity!

Solution: First, we have a sample size of 70 which is greater than or equal to 20. Additionally, the dotplot shown in Figure (1) above = namely, our sample distribution - is not strongly skewed. Therefore, the validity conditions are met.

Caution

Remember - when checking the validity conditions for a one-sample t -test, you need to check BOTH conditions!

From now on, we won't use the One Mean applet to conduct a theory-based approach; instead, we'll use the **Theory-Based Inference Applet**.

Question 6

Navigate to the **Theory-Based Inference Applet**, and:

- 1) Use the dropdown menu to select "One Mean"
- 2) Enter the sample size, sample mean, and sample standard deviation
- 3) Click "Calculate"
- 4) Check the "Test of Significance" box
- 5) Input the correct value and sign for the hypothesis
- 6) Click "Calculate"

Include a screenshot of your screen, including any/all inputs and outputs, below.

Solution:

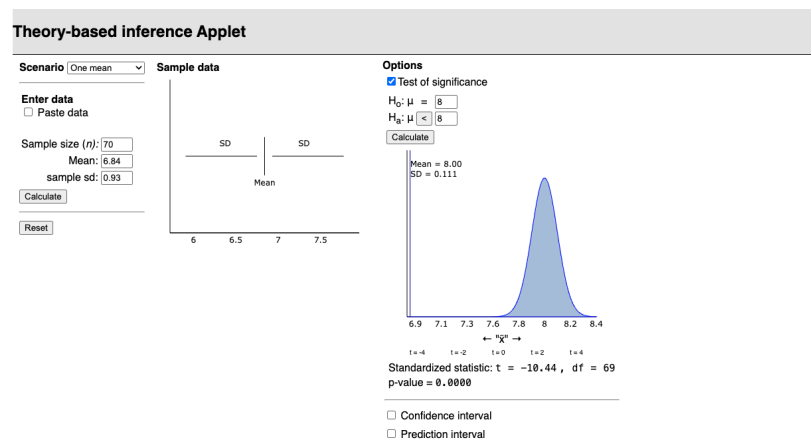


Figure 5: Screenshot from the Theory-Based Inference Applet.

Types of Errors

Reject Or Not?

I've been pretty careful to avoid using terms like "reject the null" or "fail to reject the null" up until now. That's intentional - I really do urge you to get into the habit of thinking in terms of a *sliding scale of strength of evidence against the null hypothesis*! But, in many practical cases, we need to make a determination of which - null or alternative - we think is the more plausible hypothesis.

If we conclude there to be sufficiently strong evidence against the null, we say that we **reject the null in favor of the alternative**. Otherwise, we say that we **fail to reject the null in favor of the alternative**.

⚠ Caution

Don't say "accept the null-" though some statisticians allow this wording, the vast majority (myself included) do not. When we "fail to reject," that's all we're doing - we're just saying "there wasn't enough evidence to outright reject the null." That does not logically imply that the null must be true, or that we are "accepting it."

💡 Pro Tip

For now (at this point in the course), we'll reject the null hypothesis in favor of the alternative hypothesis whenever we have "strong" or "very strong" evidence (as indicated by our various tables quantifying strength of evidence) against the null hypothesis.

Errors

Now, hypothesis testing is NOT a guarantee. This means, we could potentially make one of two mistakes:

- We could reject the null hypothesis, even though the null hypothesis was true.
 - We sometimes call this a "false alarm," because the researcher believed that they discovered an effect/difference when there actually was no effect/difference. This is called a **Type I Error**.
- We could fail to reject the null hypothesis, even though the null hypothesis was false.
 - We sometimes call this a "missed opportunity," because the researcher failed to identify an effect/difference that really was present. This is called a **Type II Error**.

⚠ Caution

To be clear - we'll never know if we committed an error, let alone which type of error we committed (if one was committed). (Can you explain why that is?)

Let's make sure you understand these concepts.

 Question 7

Fill in the entries in the following table, identifying each cell (situation) as either a “Correct decision,” “Type I Error,” or “Type II Error.”

		(What is True)	
		Null Hypothesis is True	Null Hypothesis is False
(What we decide, based on data)	Reject Null Hypothesis	Type I Error	Correct Decision
	Fail to Reject Null Hyp.	Correct Decision	Type II Error

Table 2: Error classification table.

 Question 8

Based on the completed table in Table (2) above, what type of error could we have committed in this study? Explain. **Hint:** first think about whether the data led to rejecting the null hypothesis or not. Then ask what kind of error is possible with that decision.

Solution: For such a small p -value (0.000), we’d reject the null for pretty much any cutoff we choose. So, the only type of Error we could have committed is a **Type I Error**, since a Type II error would require us to have failed to reject the null.

 Question 9

Based on your answer to Question (8) above, interpret the type of error in the context of this study.

Solution: A type I error means that the true population mean sleep time is actually 8 hours, but our sample data incorrectly led us to conclude that the mean sleep time is less than 8 hours.